

GALA: ANATOMY-AWARE LANGUAGE-MOTION ALIGNMENT FOR EFFICIENT TEXT-TO-MOTION RECTIFIED FLOW

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ABSTRACT

Text-to-motion generation requires both accurate language grounding and efficient motion synthesis. Most latent generators condition on pooled text features, so local linguistic cues are not explicitly associated with the body parts that realize them. We propose GALA, an anatomy-aware text-to-motion framework that links skeletal topology to language conditioning in a continuous generator. A skeleton-graph VAE provides global and five fixed-index body-part descriptors for alignment, while a stride-4 temporal latent is generated by a rectified-flow DiT. Learnable anatomical queries extract part-level CLIP features and align them with the corresponding motion descriptors; decoded endpoint predictions are further constrained by motion-space kinematics. On HumanML3D, GALA obtains R@3 0.806 and FID 0.272 with 20 NFEs, and reaches FID 0.224 with 50 NFEs. It also achieves R@3 0.804 and MM Dist 2.674 on KIT-ML. Ablations show that part-indexed alignment recovers retrieval performance lost under global-only alignment, while kinematic supervision further improves fidelity.

Index Terms— text-to-motion, rectified flow, skeleton graph, cross-modal alignment, human motion generation

1. INTRODUCTION

Text-to-motion (T2M) maps a sentence to a variable-length pose clip [1, 2]. HumanML3D [1] and KIT-ML [3] are standard benchmarks and are commonly evaluated under the Guo protocol using R-Precision, FID [4], multimodal distance, and Diversity over repeated evaluations [1, 2]. Diffusion on raw 263-D poses [2, 5] is faithful but slow; latent diffusion [6] and discrete tokens [7, 8, 9] push FID down at the cost of a codebook or a long denoising chain. Rectified flow [10, 11] replaces that chain with a direct ODE trajectory and enables low-NFE continuous generation.

A remaining issue is how local language cues are associated with the articulated body structure during generation. Existing latent T2M methods model motion efficiently, but the relation between token-level semantics and specific body parts is usually left implicit. GALA addresses this issue by using the skeleton topology to define a fixed anatomical indexing shared by motion descriptors and language queries. The resulting part-level language features are used directly by the rectified-flow generator, while decoded endpoint predictions are regularized in motion space. This design lets us examine separately whether errors arise from motion representation or from latent transport.

Contributions.

1. **Anatomy-aware motion tokenizer.** A CTR graph encoder plus stride-4 VAE maps 263-D poses to 256-D latents at 5 Hz, preserving skeletal dependencies and exposing part-structured features. Val reconstruction FID is 0.003.

2. **Part-indexed language-motion alignment.** Beyond pooled InfoNCE, five fixed-index queries condition the flow and are supervised by corresponding graph-pooled body parts; a collapse audit verifies that their attention profiles remain distinct.
3. **Kinematic-aware rectified flow.** An 8-block DiT learns a straight latent ODE under classifier-free guidance [12], with velocity, bone-length, and foot-contact penalties in decoded motion space, enabling 20–50 NFE sampling.

We compare with native-HumanML3D generators (MDM, MLD, M2DM, T2M-GPT, EMDM, and MoMask). GENMO [13] is included as a generalist SMPL reference.

2. RELATED WORK

Pose-space DDPMs [2, 5, 14] are faithful but slow. MLD [6] diffuses in a VAE latent—the same split we use, without a skeleton graph or a straight flow. T2M-GPT [7], M2DM [9], MoMask [8], and MMM [15] quantize motion; we stay continuous because a 256-D VAE already reconstructs at FID 0.003. Consistency models [16] and EMDM [17] cut the denoising chain; rectified flow [10, 11] learns a direct path that a DiT [18] can parameterize under CFG [12]. FlowMotion [19] predicts clean motion in conditional flow matching, MotionFlux [20] combines rectified flow with preference alignment, and MotionHiFlow [21] uses a hierarchical temporal flow with topology-aware encoding. In contrast, GALA focuses on anatomically structured conditioning: fixed query-to-part identities connect language grounding to DiT conditioning, and the decoded flow endpoint is supervised with motion-space kinematics. ST-GCN [22] and CTR-GCN [23] supply the topology prior. Part-level T2M [24, 25] generates or attends over limbs; GALA instead aligns language tokens to part latents inside one continuous flow. GENMO [13] is a generalist SMPL [26] model; its HumanML3D results rely on an SMPL-to-HumanML3D conversion that the authors identify as an inherent disadvantage, so we use it as a reference rather than a directly matched representation baseline.

3. METHOD

Fig. 1 summarizes the three-stage design. The key principle is to preserve a common anatomical indexing from the skeleton graph to language grounding and finally to flow conditioning: topology defines the body-part slots, text tokens are aligned to those slots, and the resulting global/part semantics guide the continuous transport trajectory.

Representation. A clip is $x_{1:T} \in \mathbb{R}^{T \times 263}$, $T \leq 196$, 20 fps, Guo features [1]: root motion, local joints, velocities, and four foot-contact channels. KIT-ML uses the analogous 251-D, 21-joint feature. Text is frozen CLIP ViT-B/32 [27] tokens $c_{1:N} \in \mathbb{R}^{N \times 512}$.

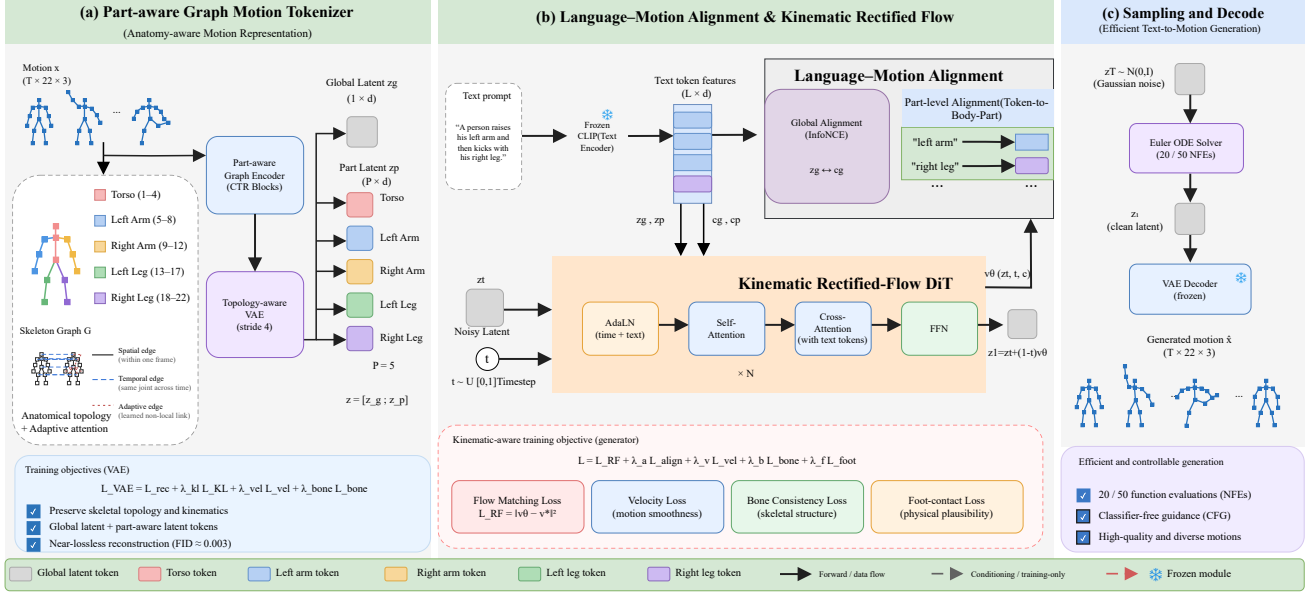


Fig. 1. GALA overview. **(a)** An anatomy-aware graph encoder models spatial, temporal, and adaptive skeletal dependencies, and topology-aware pooling exposes a global descriptor together with five fixed-index part descriptors for language alignment, while a stride-4 VAE provides the temporal latent sequence used by the generator. **(b)** Frozen CLIP token features are grounded to the same anatomical slots through global and token-to-body-part alignment; the aligned language features condition a rectified-flow DiT operating on the stride-4 temporal VAE latent sequence; its endpoint estimate is regularized by velocity, bone-consistency, and foot-contact objectives. **(c)** At inference, a 20- or 50-NFE Euler solver transports Gaussian noise to a clean latent, which the frozen VAE decoder maps to motion.

Part-aware graph tokenizer. Local joint positions and velocities form a J -node graph. Two CTR blocks combine anatomical spatial/temporal connectivity with sample-dependent adaptive relations; an ST-GCN variant (static adjacency only) and a graph-free conv VAE are the tokenizer controls. Joints are pooled into $P=5$ parts $\mathcal{P} = \{\text{torso, L-arm, R-arm, L-leg, R-leg}\}$, giving a global graph descriptor z_g and ordered part descriptors $z_p \in \mathbb{R}^{P \times d}$ used by the alignment branch. These descriptors are conditioning/alignment features rather than the temporal state transported by rectified flow. Separately, a 1-D conv VAE [28] with stride 4 emits the temporal posterior parameters $\mu, \log \sigma^2 \in \mathbb{R}^{L \times 256}$, $L = \lceil T/4 \rceil$; the sampled/mean VAE latent sequence is the state used by the generator. Stage-1 training is

$$\mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{rec}} + \lambda_{\text{KL}} \mathcal{L}_{\text{KL}} + \lambda_{\text{vel}} \mathcal{L}_{\text{vel}} + \lambda_{\text{bone}} \mathcal{L}_{\text{bone}}. \quad (1)$$

After 35k AdamW steps the tokenizer is frozen.

Fine-grained language-motion alignment. As illustrated in Fig. 1(b), GALA retains global sentence-motion alignment while introducing fixed anatomical slots for local semantics such as “left arm” and “right leg.” Global alignment uses symmetric InfoNCE on pooled CLIP and pooled latents, $\mathcal{L}_{\text{global}} = \text{InfoNCE}(\bar{c}, \bar{z})$, weight 0.25. On top of that, P learnable anatomical queries q_p attend to CLIP tokens,

$$\tilde{c}_p = \text{Attn}(q_p, c_{1:N}), \quad (2)$$

and are trained to match the corresponding part latent

$$\mathcal{L}_{\text{part}} = \frac{1}{P} \sum_{p=1}^P \text{SymNCE}(\{z_{b,p}\}_{b=1}^B, \{\tilde{c}_{b,p}\}_{b=1}^B), \quad (3)$$

where negatives are other examples in the minibatch. Each query is contrasted only with the graph-pooled latent at the same ordered

index: torso, left arm, right arm, left leg, and right leg. This predefined correspondence prevents permutation ambiguity among the five slots. The same $\tilde{c}_p$ are concatenated to CLIP tokens as extra DiT context, so the part branch remains active during sampling rather than serving only as a projection head. The anatomical correspondence is supervised by the fixed-index contrastive targets rather than inferred from the attention weights.

Kinematic rectified flow. Given clean μ (detached), $t \sim \mathcal{U}(0, 1)$, $\varepsilon \sim \mathcal{N}(0, I)$,

$$z_t = (1 - t)\varepsilon + t\mu, \quad v^* = \mu - \varepsilon. \quad (4)$$

An 8-block DiT (width 512, 8 heads) predicts $v_\theta(z_t, t, c)$ with AdaLN modulation on the pooled text vector, self-attention on latent tokens, and cross-attention to CLIP+ part tokens. Condition dropout 0.1 enables CFG [12]. The flow loss is $\mathcal{L}_{\text{RF}} = \|v_\theta - v^*\|_2^2$. We additionally decode the endpoint estimate $\hat{z}_1 = z_t + (1 - t)v_\theta$ with the frozen decoder and penalize motion-space kinematics:

$$\mathcal{L}_{\text{vel}} = \frac{1}{T-1} \sum_t \|(\hat{x}_{t+1} - \hat{x}_t) - (x_{t+1} - x_t)\|_1, \quad (5)$$

$$\mathcal{L}_{\text{bone}} = \sum_{(i,j) \in E} \|\hat{p}_i - \hat{p}_j\|_2 - \|p_i - p_j\|_2, \quad (6)$$

plus an ℓ_1 term $\mathcal{L}_{\text{foot}}$ on the four foot-contact channels. The generator objective is

$$\mathcal{L} = \mathcal{L}_{\text{RF}} + \lambda_g \mathcal{L}_{\text{global}} + \lambda_p \mathcal{L}_{\text{part}} + \lambda_v \mathcal{L}_{\text{vel}} + \lambda_b \mathcal{L}_{\text{bone}} + \lambda_f \mathcal{L}_{\text{foot}}. \quad (7)$$

Sampling is Euler, $z \leftarrow z + \frac{1}{N}(v_\theta + \omega(v_c - v_\theta))$, with $N \in \{10, 20, 50\}$ and $\omega \in \{2.0, 2.5\}$.

Training. Stage 1: VAE only, 35k steps, effective batch 16, bf16. Stage 2: freeze graph+VAE, train flow+alignment, 100k steps, lr 2×10^{-4} , EMA 0.9999. The graph+global control in Table 1 uses the same tokenizer and DiT budget as the full model. Full GALA adds part alignment and decoded-space kinematic supervision.

4. EXPERIMENTS

Protocol. Official HumanML3D test [1], 4544 clips, Guo / MDM evaluator [2], batch 32, 20 replications, mean $\pm$ 95% CI. Diversity is compared to real motion (9.503), not maximized. KIT-ML uses the same evaluator on the 251-D official split. Hardware: RTX 4060 8 GB. GALA has 59.9M parameters (DiT 53.6M, VAE 3.2M, graph 1.6M).

Literature rows in Table 1 for T2M / MDM / M2DM / EMDM / GENMO follow GENMO Table 4 [13]; MLD / T2M-GPT / MoMask follow MoMask [8]. Because GENMO converts SMPL motion to the HumanML3D representation, its FID is reported as a reference rather than treated as a directly matched comparison.

4.1. HumanML3D

Table 1 summarizes the HumanML3D comparison.

Semantic metrics at low NFE. At 20 NFes GALA reaches R@3 $0.806 \pm .002$ and MM Dist $3.061 \pm .009$, improving the graph+global control by 0.038 and 0.187, respectively. Its R@3 is comparable to the reported 0.807 of MoMask while using a continuous 20-step trajectory. R@1 / R@2 are $0.494 \pm .003$ / $0.701 \pm .002$; Diversity 9.69 remains close to Real 9.50.

Motion fidelity. At 50 NFes, GALA obtains an FID of 0.224, improving over the graph+global control (0.312) and M2DM (0.352), while remaining behind T2M-GPT / EMDM / MoMask (0.141 / 0.112 / 0.045). The tokenizer itself reconstructs the validation set at FID 0.003. The much larger FID after generation therefore suggests that the main fidelity loss is introduced by the latent transport process rather than by motion reconstruction. GENMO reports FID 0.216 after SMPL-to-HumanML3D conversion; because this representation path differs from native HumanML3D generation, we treat that number as contextual reference.

4.2. KIT-ML

Table 2 uses the same Guo protocol on the official KIT-ML split. Literature numbers are from MoMask [8] and MDM [2]. GALA uses the 21-joint graph and otherwise follows the HumanML3D training recipe. It obtains R@3 0.804 and MM Dist 2.674, compared with MoMask’s 0.781 and 2.779, respectively, while its FID remains higher (0.402 vs. 0.204). The same pattern observed on HumanML3D—stronger retrieval with a remaining fidelity gap—indicates that the effect of part-level language grounding is not limited to one dataset.

Table 2. KIT-ML test under the Guo protocol (20 replications, 20 NFes).

Method	R@3 $\uparrow$	FID $\downarrow$	MM Dist $\downarrow$	Diversity $\rightarrow$
Real	0.779	0.031	2.788	11.08
T2M [1]	0.681	3.022	3.488	–
MDM [2]	0.396	0.497	9.191	10.85
MLD [6]	0.734	0.404	3.204	–
T2M-GPT [7]	0.745	0.514	3.007	–
MoMask [8]	0.781	0.204	2.779	–
GALA	0.804 $\pm$.006	0.402 $\pm$.020	2.674 $\pm$.016	11.15 $\pm$.11

4.3. What each module contributes

We isolate the contribution of each component under a controlled setting. Table 3 starts from a rectified-flow baseline with a convolutional tokenizer and no contrastive alignment, then successively adds graph encoding, global alignment, part-indexed alignment, and kinematic supervision. Every row uses the same training budget, latent width, DiT, optimizer, and official 20-replication test protocol.

Table 3. HumanML3D component ablation (20 NFes, CFG 2.5; 20-rep test).

Model	Graph	Glob.	Part	Kin.	R@3 $\uparrow$	FID $\downarrow$
Base RF					0.789	0.402
+ Graph	✓				0.797	0.369
+ Global	✓	✓			0.769	0.337
+ Part	✓	✓	✓		0.801	0.302
GALA	✓	✓	✓	✓	0.806	0.272

Table 1. HumanML3D test, Guo protocol, 20 replications on 4544 clips (mean $\pm$ 95% CI). $\uparrow$ higher, $\downarrow$ lower, $\rightarrow$ closer to Real.

Method	Rep.	NFE $\downarrow$	R@3 $\uparrow$	FID $\downarrow$	MM Dist $\downarrow$	Diversity $\rightarrow$
Real [1]	HML3D	–	0.797	0.002	2.974	9.503
T2M [1]	HML3D	–	0.740	1.067	3.340	9.188
MDM [2]	HML3D	1000	0.611	0.544	5.566	9.559
MLD [6]	HML3D	50	0.772	0.473	3.196	9.724
M2DM [9]	HML3D	–	0.763	0.352	3.134	9.926
T2M-GPT [7]	HML3D	–	0.775	0.141	3.121	9.761
EMDM [17]	HML3D	10	0.786	0.112	3.110	9.551
MoMask [8]	HML3D	–	0.807	0.045	2.958	–
GENMO [13]	SMPL	–	0.632	0.216	3.466	11.342
Graph+Global, CFG 2.5	HML3D	20	0.768 $\pm$.002	0.330 $\pm$.011	3.248 $\pm$.008	9.529 $\pm$.085
Graph+Global, CFG 2.0	HML3D	50	0.749 $\pm$.003	0.312 $\pm$.009	3.342 $\pm$.009	9.365 $\pm$.115
GALA, CFG 2.5	HML3D	20	0.806 $\pm$.002	0.272 $\pm$.007	3.061 $\pm$.009	9.692 $\pm$.087
GALA, CFG 2.0	HML3D	50	0.793 $\pm$.003	0.224 $\pm$.006	3.107 $\pm$.008	9.643 $\pm$.082

The graph tokenizer reduces FID from 0.402 to 0.369. Interestingly, adding global alignment further lowers FID to 0.337 but decreases R@3 from 0.797 to 0.769. This non-monotonic result suggests that sentence-level alignment alone can improve the generated distribution without preserving localized text–motion correspondence. After part-indexed alignment is introduced, R@3 recovers to 0.801 and FID improves to 0.302. Kinematic supervision gives the final 0.806 R@3 and 0.272 FID. The ablation therefore attributes the retrieval recovery mainly to explicit part-level grounding, while the decoded-space constraints provide an additional fidelity gain.

4.4. Tokenizer: graph vs. conv

Table 4. Tokenizer reconstruction on HumanML3D val (1504 clips).

Tokenizer	R@3 ↑	Recon FID ↓	MPJPE ↓	Bone ↓	Vel. ↓
Conv-VAE	0.755	0.012	0.105	0.061	0.039
ST-GCN-VAE	0.748	0.003	0.088	0.050	0.038
CTR-Graph-VAE	0.750	0.003	0.080	0.051	0.038

Table 4 compares equally trained conv, static ST-GCN, and adaptive CTR tokenizers. Graph encoding cuts FID from 0.012 to 0.003 and MPJPE from 0.105 to 0.080, while R@3 is unchanged. ST-GCN already captures the FID gain; CTR’s distinct benefit is lower MPJPE (0.088→0.080).

4.5. Sampling and the quality–efficiency Pareto

Table 5. Validation sampling scan (1504 clips, one seed).

Steps	CFG	FID ↓	R@3 ↑	MM Dist ↓
10	1.0	1.233	0.628	4.062
10	2.5	0.382	0.771	3.204
20	2.5	0.335	0.781	3.164
50	2.0	0.290	0.753	3.301
50	2.5	0.322	0.767	3.177

The val scan (Table 5) exposes a semantic–fidelity trade-off: 20 steps / CFG 2.5 favors R@3, whereas 50 / 2.0 favors FID. For single-sample inference on an RTX 4060 (bf16, batch size 1, $T=196$), 10/20/50 NFE require 70/140/340 ms per clip (10 timed runs after warm-up), corresponding to 14.4/7.1/2.9 clips/s.

5. CONCLUSION

GALA links skeletal topology, part-level language grounding, and rectified-flow generation through a shared anatomical indexing. Experiments on HumanML3D and KIT-ML show that fixed-index part alignment improves text–motion retrieval, while decoded-space kinematic supervision further improves motion fidelity. The graph tokenizer already provides high-quality reconstruction, whereas generated FID remains noticeably higher. This gap suggests that future improvement should focus primarily on the low-step latent transport model rather than on increasing tokenizer capacity.

6. REFERENCES

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